

CATALAN NUMBERS

MATH 300

Choose one of the following problems and find some initial data.

Problem 1: Let s_n be the number of n -step staircases that you can construct with n -rectangles? Find s_1 , s_2 , s_3 , and s_4 by drawing pictures. See the reverse side for the $n = 5$ case.

Problem 2: Let c_n be the number of up-right paths on an $n \times n$ grid that do not pass above the diagonal? Find c_1 , c_2 , c_3 , and c_4 by drawing pictures. See the reverse side for the $n = 5$ case.

